

Bug Deduplicator

AI-assisted duplicate-defect triage for issue trackers

The problem

In an active project, the same defect gets reported more than once — by different people, in different words, days apart. Triagers waste time asking "have we seen this before?" and answering it by hand. Missed duplicates fragment the picture of a single underlying fault across several tickets.

Why keyword search isn't enough

Two genuine duplicates can share almost no words (one reporter writes "reboots after update," another "goes offline post-firmware-push"), while two unrelated bugs can look nearly identical. Matching on text alone fails in both directions. The useful signal is **root cause**, not vocabulary — which is exactly what a language model can reason about.

How it works

A two-stage pipeline: cheap retrieval narrows the field, then model judgment decides. Retrieval keeps recall high; the model keeps precision high by rating each candidate on subsystem and root cause and staying deliberately conservative.

1 New defect A bug is filed in the tracker and triggers the agent.	2 Retrieve A scoped query pulls candidate defects from the same project (recall).	3 Judge The model rates each candidate on root cause, not wording (precision).	4 Suggest A confidence-scored comment is posted; a human confirms.
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Worked example

Incoming: "Device shows offline on the dashboard while readings keep arriving in the log — only the status indicator is wrong."

ATLAS-354 possible duplicate · medium-high

Same subsystem and root cause — the status flag and the telemetry pipeline update from different sources. Shares almost no wording with the incoming report, so keyword search would miss it.

Design principles

Conservative

Flags 'related' or 'possible' unless the fault is clearly identical — false alarms erode trust.

Silent when clear

Posts nothing when no real match exists — no noise on the board.

Human-in-the-loop

Suggests only. Never links or closes; a triager decides.

Scoped & owned

Runs per-project on a schedule; no admin dependency, low running cost.